



Continuous Quality Improvement principles and tools

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Quality

**is doing the right thing,
on the right time within
the right way to achieve
best possible results.**

Dimensions of Health care Quality





1. QUALITY ASSURANCE (QA):



A proactive process focused on ensuring that healthcare services meet defined quality standards before problems occur.

Goal: prevent errors by establishing systems, protocols, and standards.

2. Quality Control (QC)

- A reactive process used to identify and correct issues in healthcare delivery after they occur.
- Goal: Detect and fix deviations from quality standards.

3. Quality Improvement (QI)

- An ongoing, systematic process to analyze performance and make measurable improvements in healthcare processes and outcomes.
- Goal: Continuously enhance care delivery and patient outcomes. All of them are called Total Quality Management

Principles of Quality Improvement in Healthcare

Patient-centered care:

Focus on patient needs, preferences, and values.

Continuous

improvement: Regularly assess and improve care processes.

Collaboration:

Foster teamwork among healthcare professionals.

Evidence-based practice:

Use the best available research evidence to guide care.

Transparency:

Share data, results, and lessons learned.

Quality Improvement Process:

- Ensuring a culture that establishes quality in every practice
- Determining and prioritizing areas of potential improvement
- Data collection and analysis
- Communicate with your acquaintances about the results
- A commitment to a continuous evaluation
- Expand your successes



Role of Data Collection and Analysis in Quality Improvement



1. Identify areas for improvement:
analyze data to pinpoint gaps in care.



2. Set goals and objectives: use data to establish realistic targets.



3. Monitor progress: track data to assess effectiveness of interventions.



4. Evaluate outcomes: analyze data to determine impact on patient care.

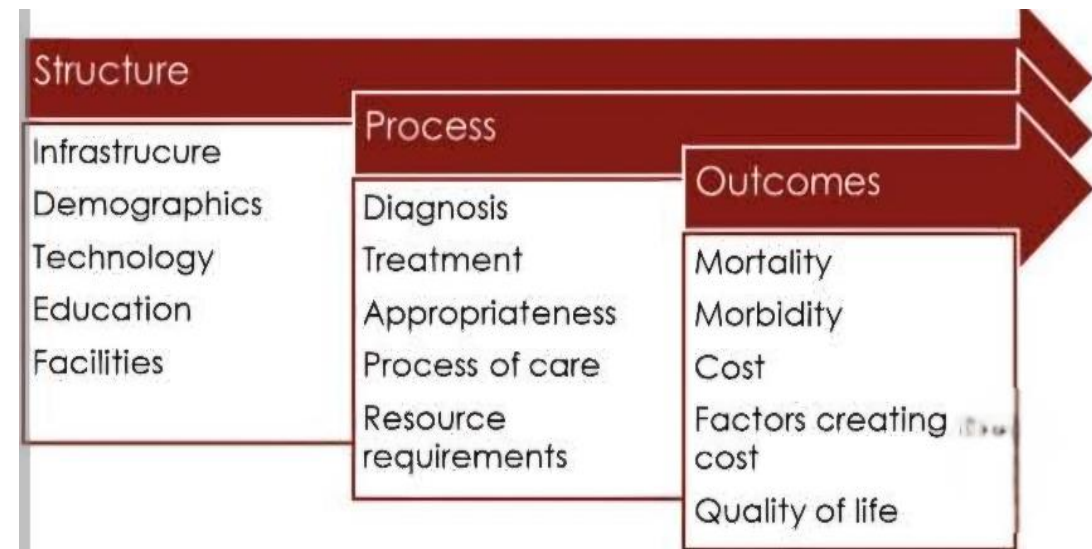


5. Inform decision-making: use data to guide policy and practice changes.



Quality Improvement Models

- **Donabedian model:**
Focus on structure, process, and outcomes



AI-Enhanced Donabedian's
...Framework: A 21st-Century

Six Sigma model

- **Defining**

When starting out, the first thing to do is to determine the goals of the client. Ask yourself what it is you are trying to achieve or what expectations or internal goals, for instance.

- **Measuring**

The next step to do is measure data from the specific process in order to understand the current performance level and input information to that process.

- **Analyzing**

Review or analyze those processes and review data then try to build to inefficiencies along the way.

- **Improving**

In order to improve these processes, act in accordance with what the data suggests. With the information, find improvement solutions and implement them.

- **Controlling**

Finally, check for problems with the solutions, see that a level of quality control is maintained. If there are no issues, standardize procedures as a basis of the process.



Quality Improvement Tools

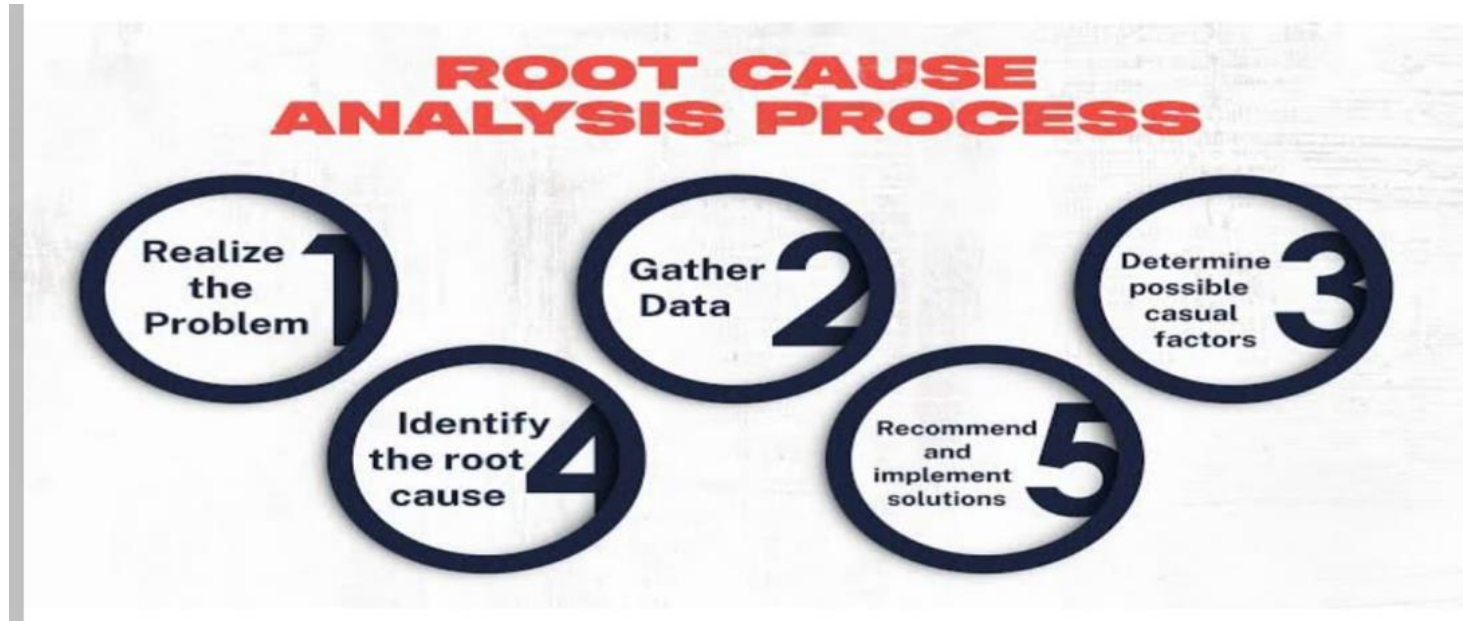
- **Quality Improvement Tools:** is the process that helps better understanding, analyzing and communicate, quality improvement efforts.
- **Root cause analysis (RCA).** (Fishbone diagram, 5 Whys Technique, Cause and Effect Diagram).
- **Failure mode and effects analysis (FMEA).**
- **SWOT Analysis.**
- **Flowcharts.**
- **Control charts.**
- **Pareto chart.**
- **Run charts.**
- **Decision Tree Analysis.**



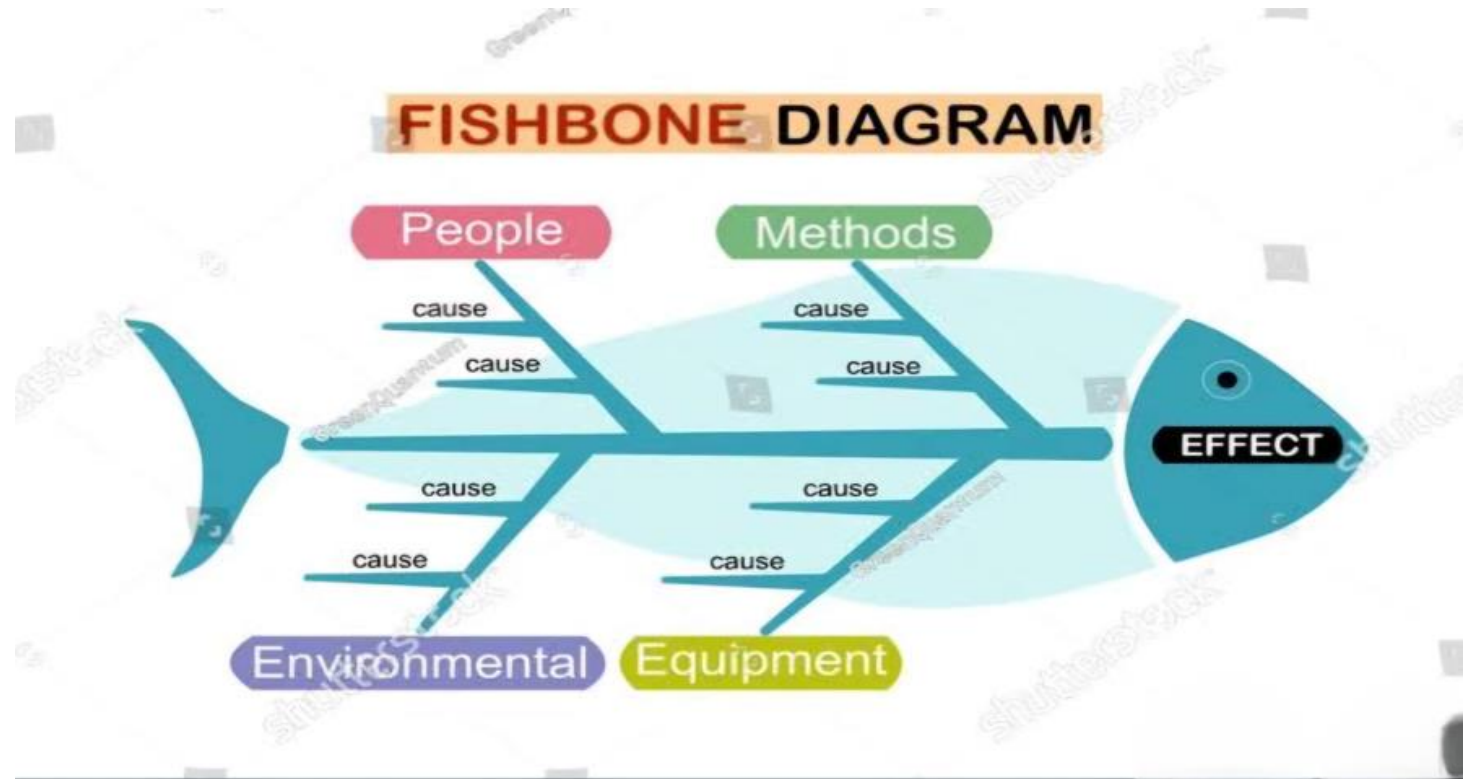
1. Root cause analysis (RCA):

Identify underlying **causes** of problems.

- Realize the Problem
- Gather Data
- Determine possible causal factors
- Identify the root cause
- Recommend and implement solutions



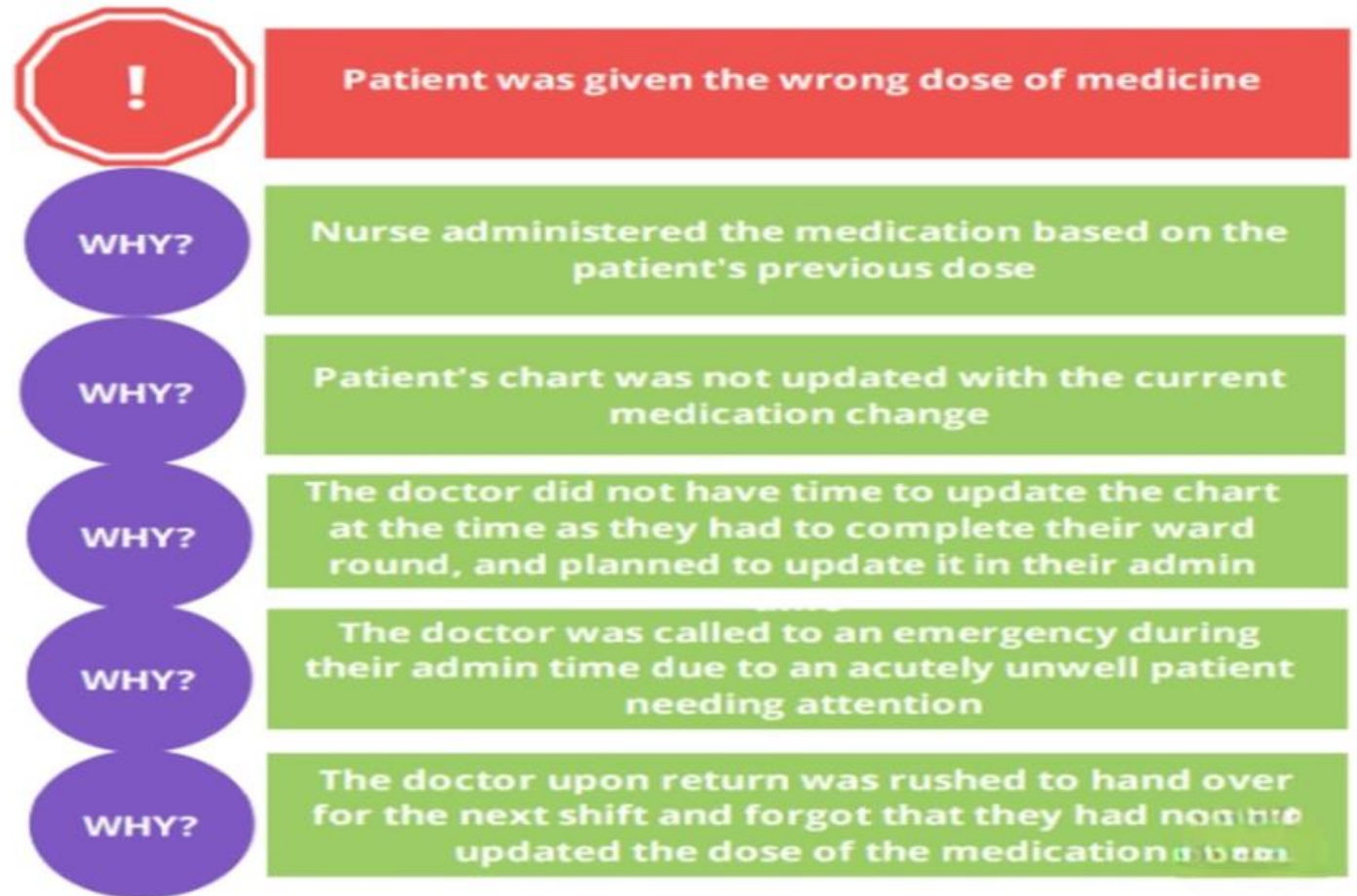
1. Root cause analysis (RCA):



- **A.Fishbone diagram:** Visualize **potential causes** of problems.

1. Root cause analysis (RCA):

- B. 5 Whys Technique : depends on Asking Why 5 times to **Identify the root cause of the problem.**



Failure mode and effects analysis (FMEA):

Identify potential failures and their impact.

The 7 Steps of PFMEA

PFMEA is conducted when the following occurs in the organization:

- New changes are made on existing processes;
- Introduction to new technology, equipment, or process steps;
- Moving a process to a new facility, or department.

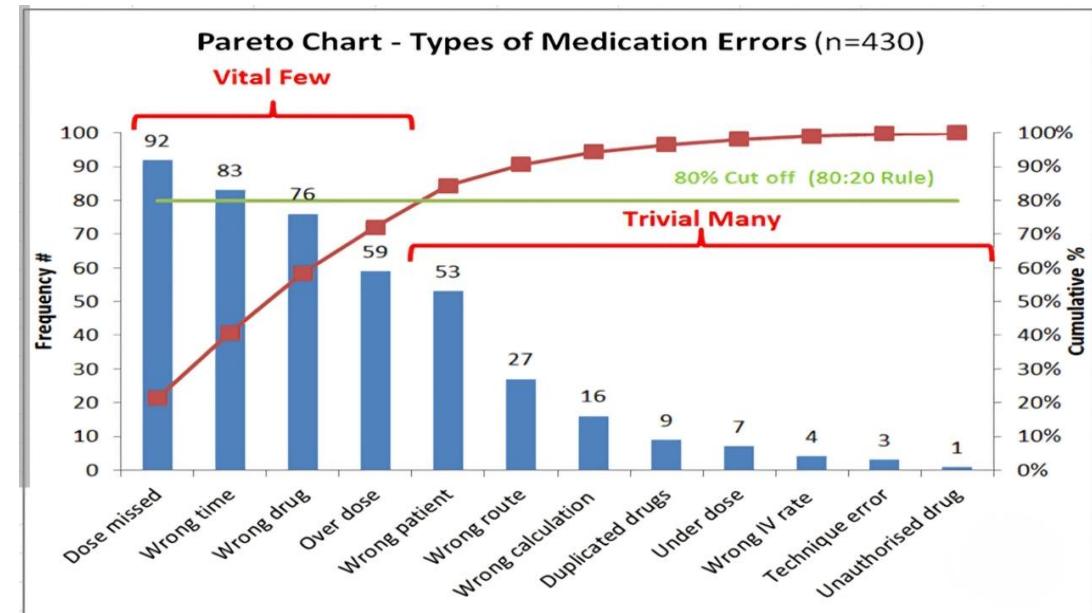


SWOT
analysis:
Identify
strengths,
weaknesses,
opportunities,
and threats.



4. Pareto chart: graph that helps identify the most significant factors contributing to a problem.

- It is based on the Pareto Principle (80/20 rule) — meaning that 80% of problems often come from 20% of causes.



Another tools

- **5. Flowcharts:** Map care processes to identify inefficiencies.
- **6. Control charts:** Monitor data to detect trends and anomalies.
- **7. Run charts:** Visualize data to track progress over time.
- **8. Shewhart chart.**



Challenges and Opportunities for quality improvement:

- **1. Data quality and availability:** Ensure accurate, reliable data.
- **2. Stakeholder engagement:** Involve patients, families, healthcare professionals.
- **3. Sustainability:** Ensure long-term viability of quality improvement initiatives.
- **4. Spread and scale:** النشر والتوسّع
Disseminate successful initiatives across organization.

